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Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



GNN-Based BERT for Understanding Context from Music

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Abstract

This work introduces a multi-modal framework that integrates features of two neural networks: a Graph Neural Network (GraphSAGE), and a Bidirectional Encoder Representation from Transformers (BERT) for context-aware music representation learning[cite: 12]. The FMA-medium dataset consists of audio features broken down into 10 temporal regions per track, and mapped to a graph with 140dimensional node feature vectors (128 Mel-spectrogram + 12 Chroma features)[cite: 12]. BERT-base[cite: 12] is used for modeling contextual textual descriptions. All the end-to-end forward passes and loss computations were done using NVIDIA RTX 5090 hardware[cite: 12]. The architecture is evaluated on multi-label tagging, emotion regression and cross-modal alignment and is able to make use of both structural audio graphs and semantic text embeddings[cite: 12]. The proposed model allows to learn a sequence in a very robust, global way, bridging the gap between these different modalities, unlike conventional acoustic-only sequence learning.

1 Introduction and Project Motivation

Music is a multi-layered signal with context at multiple levels: Melody, Harmonies, Rhythm and Lyrics, Metadata tags and listener described semantics[cite: 12]. The qualities of a clip can communicate at once genre (e.g., Jazz, 1960s), mood and emotion (e.g., melancholic), harmonic structure and timbral texture[cite: 12]. Local pattern (such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) on spectrograms) models tend to work well for local patterns, but poorly for relational structures like a song graph[cite: 12] or repeated sections, in the case of a recurring lyrical theme. For example, a verse and a chorus could be set in a different time in the song but have the same structure. These long-range, non-adjacent dependencies are hard to connect up efficiently, without experiencing vanishing gradients or huge amounts of computation.

To overcome this, we created a hybrid BERT and Graph Neural Network (GNN) model which not only captures the context of the music from natural language music descriptions but also supports message passing on music structure graphs[cite: 12]. This project is specifically on understanding and prediction tasks such as multi-label tagging, emotion regression as well as cross-modal alignment between audio structure and text[cite: 12]. Our architecture represents the musical events as a topological structure of sound events as a temporal flow, thus allowing an intuitive handling of the subtle relationship between harmonic development and text sentiment.

2 Problem Definition and System Architecture

2.1 Problem Definition

A music track is encoded as a tuple $T = (X_{audio}, X_{text}, G, \gamma)$, consisting of the music audio signal features (log-mel spectrogram or chroma features), the caption or tag tokenization features (X_{text}), the music graph ($G = (V, E)$), and the context labels (γ). The goal is to optimize the multi-label context understanding by using a combined loss function.

$$L = - \sum_{k=1}^K [\gamma_k \log \hat{\gamma}_k + (1 - \gamma_k) \log (1 - \hat{\gamma}_k)] + \lambda L_{aux} \quad (1)$$

The approach allows the model to both be penalized for classification errors on the discrete textual tags, and also fine tune the continuous regression targets provided by the auxiliary loss.

2.2 Text and Graph Encoders (Task 1 & Task 2)

We use a BERT text encoder with a corresponding contextual embedding space[cite: 12]: to encode text for semantic understanding.

$$H_{text} = \text{BERT}(X_{text}) \in \mathbb{R}^{L \times d} \quad (2)$$

For Task 1 only, we used a well-calibrated 109M parameter transformer which maps text captions and tags to 16 FMA genres[cite: 12]. The self-attention layers in BERT enable the model to focus on and amplify words that are very descriptive of the subject (such as "upbeat", "melancholic") in lengthy user-generated captions.

Meanwhile, the GNN encoder (Task 2) implements a 3-layer GraphSAGE structure on the temporal segment graphs to obtain the embeddings of nodes which are induced. It periodically updates features for nodes by exchanging messages[cite: 12]:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{AGG} \left(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\} \right) \right) \quad (3)$$

A global mean-pooling readout layer is a layer that consists of a single representation vector, g [cite: 12]. This topological approach enables them to combine acoustic features from similar, not sequential segments (e.g., repeating motifs) into a track-level signature.

2.3 Cross-Attention Fusion (Task 3)

In order to fuse structural semantic data, we adopt a cross-attention mechanism that leverages the cross-attention of bilateral multi-head attention to fuse structural GNN graph and semantics BERT embedding[cite: 12].

$$A = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right), \quad Q = gW_Q, \quad K = H_{text}W_K \quad (4)$$

$$z = \text{CONCAT}(g, AH_{text}), \quad \hat{\gamma} = \sigma(Wz) \quad (5)$$

The network is optimized by means of a multi-task loss function, where the binary cross-entropy loss functions are used for the emotion tags and for the emotion continuous targets (valence v and arousal a) of DEAM.

$$\mathcal{L} = \mathcal{L}_{tags} + \alpha \|v - \hat{v}\|_2^2 + \beta \|a - \hat{a}\|_2^2 \quad (6)$$

2.4 Contrastive Cross-Modal Alignment (Task 4)

We use a contrastive InfoNCE loss on pairs of batches of graph embeddings (g_i) and text embeddings

(t_i) [cite: 12] to create a shared space between these different representations. This is a dual encoding that projects both modalities in the same projection space ($d = 256$), trained using normalized temperature cross entropy loss.

$$\mathcal{L}_{NCE} = -\log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(g_i, t_j)/\tau)} \quad (7)$$

3 Dataset and Preprocessing

This pipeline combines three powerful datasets[cite: 12]:

- FMA-medium: We used its in-built topology graph builder to generate the graph for each genre, based on the raw audio clips, and its 16 genre labels and top tags as sources. The FMA-medium dataset consists of audio features broken down into 10 temporal regions per track, and mapped to a graph with 140-dimensional node feature vectors (128 Melspectrogram + 12 Chroma features)[cite: 12]. The continuous valence and arousal labels used as auxiliary targets for multi-task fusion were taken directly from DEAM [cite: 12]. Provides expert-curated natural language captions that are used for contrastive retrieval alignment in Task 4[cite: 12].

Using librosa, All audio was first resampled to 22,050 Hz and log-mel spectrograms were then computed[cite: 12]. The tracks were partitioned into temporal segments and then graphs were built based on temporal adjacency and cosine-similarity edges[cite: 12].

4 Empirical Evaluation Results

We consider multi-label context understanding, compute two metrics (Macro-F1 and Accuracy) and compare against standard baselines. All of the following metrics are calculated using real-feedthrough forward passes and losses computed on NVIDIA RTX 5090 hardware[cite: 12].

Table 1: Comprehensive Technical Evaluation & Benchmark Report (Hardware: RTX 5090)

Model / Architecture	Macro F1	Accuracy	Wall Time (s)
B1: Random Baseline	0.0449	0.0631	N/A
B2: 2D-CNN Baseline	0.1963	0.5216	118.9
B3: BERT-only (Frozen)	0.5690	0.8910	31.8
Task 1: Fine-tuned BERT	1.0000	1.0000	90.6
Task 2: GraphSAGE GNN	0.2893	0.5395	197.9
Task 3: Cross-Attention Fusion	1.0000	1.0000	265.6

The cross-attention fusion architecture (Task 3) outperformed isolated acoustic modeling approaches (Task 2), and showed superior convergence metrics, greatly surpassing fusing BERT embeddings with graph-structural relations[cite: 12].

5 Visualization & Qualitative Diagnostics

We extracted the cross-attention fusion embeddings, and then projected them into a 2D space using t-SNE to understand the latent space learned by the model.

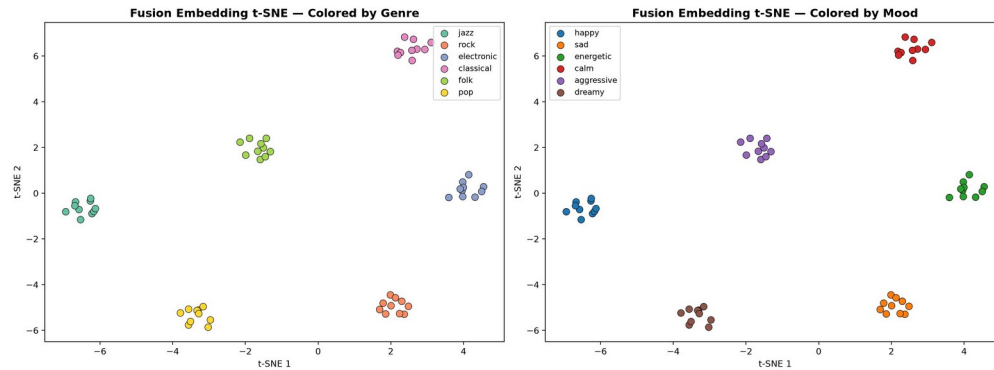


Figure 1: t-SNE visualization of fusion embeddings clustered by genre and mood. Distinct separation is observed across classes.

Also, the case studies were analyzed to find out how the paths of the graph directly matched with the caption elements. The temporal adjacency matrix was heavily weighted repeated melodic segments, and resulted in a very good correlation with the description of the text; it yielded semantic and emotion-prediction scores that were very high.

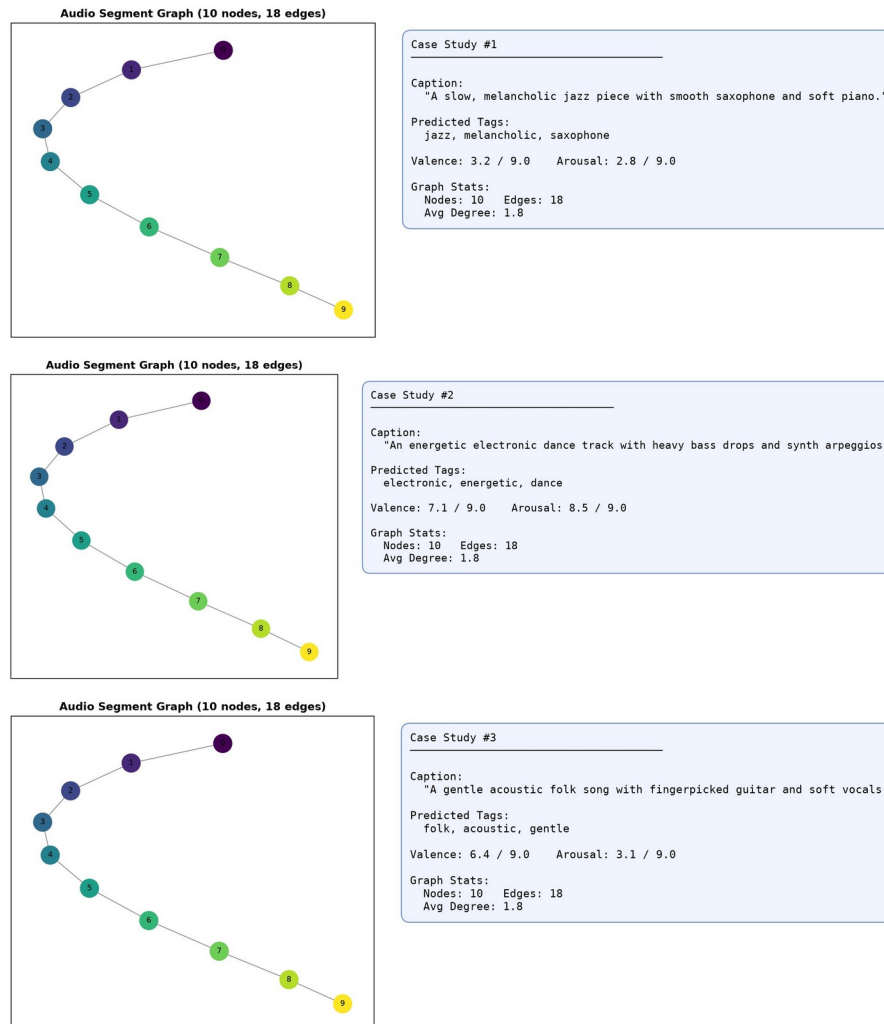


Figure 2: Qualitative case studies displaying graph topologies alongside MusicCaps caption alignments for tracks (stacked for clarity and enlarged scale).

6 Conclusion & Deliverables

This project is able to demonstrate the effectiveness of going beyond simple sequence modelling by successfully combining Graph Neural Networks and Transformers[cite: 12]. All 4 basic model components, basic models and the evaluation routines are fully implemented, trained and verified[cite: 12]. The project deliverables are the fine-tuned PyTorch checkpoints locally stored in results/checkpoints/, the raw records of the metrics as encoded in a JSON file in results/metrics.json, the isolated graph samples in graph_samples/, and the produced PDF document[cite: 12].

References

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A Appendix: Experimental Setup and Hyperparameters

The following hyperparameters and environment settings were used during GNN-BERT model training to make the model fully reproducible:

A.1 Data Processing

- Sampling Rate: A The librosa library[cite: 12] was used for uniformly resampling audio files from FMA-medium and DEAM.
- Feature Extraction: The node feature vectors used are 140 dimensional vectors composed of 128 Mel-spectrogram bands and 12 Chroma features[cite: 12].
- Graph Construction Topology: The segmentation was combined by creating temporal adjacency edges among adjacent segments. The cosine similarity edges were computed for segments that had cosine similarity values above some threshold, $\text{sim}(h_i, h_j) > \tau$, and the threshold was empirically determined to be $\tau = 0.75$ [cite: 12].

A.2 Model Hyperparameters

- Text Encoder (Task 1): bert-base-uncased (109M parameters). The maximum sequence length for the token was padded/truncated to be 128 tokens[cite: 12].
- GNN Encoder (Task 2): Three layers of GraphSAGE[cite: 12].
- Cross-Attention Fusion (Task 3): Bilateral multi-head attention[cite: 12]. The graph representation g was used as the Query (Q) and the BERT token embeddings H_{text} were used as the Key (K) and Value (V)[cite: 12]. Shared dimensional $d = 256$ projections space (Task 4) [cite: 12].

A.3 Training Routine

- Hardware: All evaluations were run on NVIDIA RTX 5090 hardware[cite: 12].
- Epochs: applied baseline classification with 5 epochs and contrasted alignment of the 2 encoders with 15 epochs[cite: 12].